

1)

```
D4_99207_OM> SELECT ODATE,COUNT(DISTINCT SNUM)
-> FROM Orders
-> GROUP BY ODATE;
+-----+-----+
| ODATE | COUNT(DISTINCT SNUM) |
+-----+-----+
| 1990-10-03 | 4 |
| 1990-10-04 | 3 |
+-----+-----+
2 rows in set (0.02 sec)
```

2)

```
D4_99207_OM> SELECT CONCAT('For the city ', CITY , ' The highest rating is: ' ,MAX(rating)) as RATING
-> FROM Customers
-> GROUP BY city;
+-----+
| RATING |
+-----+
| For the city London The highest rating is: 100 |
| For the city Rome The highest rating is: 200 |
| For the city San Jose The highest rating is: 300 |
| For the city Berlin The highest rating is: 300 |
+-----+
4 rows in set (0.00 sec)
```

3)

```
D4_99207_OM> SELECT odate , SUM(amt) AS 'total order amount'
-> FROM orders
-> GROUP BY odate
-> ORDER BY SUM(amt) DESC;
+-----+-----+
| odate | total order amount |
+-----+-----+
| 1990-10-04 | 16713.81 |
| 1990-10-03 | 8944.59 |
+-----+-----+
2 rows in set (0.00 sec)
```

4)

```
D4_99207_OM> SELECT snum , SUM(AMT) AS Total_amount
-> FROM orders
-> GROUP BY snum
-> HAVING Total_amount > 2565.84;
+-----+-----+
| snum | Total_amount |
+-----+-----+
| 1001 | 15382.07 |
| 1002 | 5546.15 |
+-----+-----+
2 rows in set (0.03 sec)
```

5)

```
D4_99207_OM> SELECT city,MAX(rating)
-> FROM Customers
-> GROUP BY city;
+-----+-----+
| city | MAX(rating) |
+-----+-----+
| London | 100 |
| Rome | 200 |
| San Jose | 300 |
| Berlin | 300 |
+-----+-----+
4 rows in set (0.00 sec)
```

6)

```
D4_99207_OM> SELECT snum,MAX(amt)
-> FROM orders
-> GROUP BY snum
-> HAVING MAX(amt)>3000;
+-----+-----+
| snum | MAX(amt) |
+-----+-----+
| 1001 | 9891.88 |
| 1002 | 5160.45 |
+-----+-----+
2 rows in set (0.03 sec)
```

7)

```
D4_99207_OM> SELECT cnum,MIN(amt)
-> FROM orders
-> GROUP BY cnum
-> ORDER BY cnum;
+-----+-----+
| cnum | MIN(amt) |
+-----+-----+
| 2001 | 767.19 |
| 2002 | 1713.23 |
| 2003 | 5160.45 |
| 2004 | 75.75 |
| 2006 | 4723.00 |
| 2007 | 1900.10 |
| 2008 | 18.69 |
+-----+-----+
7 rows in set (0.00 sec)
```

PART 2:

1)

```
D4_99207_OM> SELECT manager_id , COUNT(employee_ID)
-> FROM employees
-> GROUP BY manager_ID;
+-----+-----+
| manager_id | COUNT(employee_ID) |
+-----+-----+
| NULL | 1 |
| 100 | 14 |
| 101 | 5 |
| 102 | 1 |
| 103 | 4 |
| 108 | 5 |
| 114 | 5 |
| 120 | 8 |
| 121 | 8 |
| 122 | 8 |
| 123 | 8 |
| 124 | 8 |
| 145 | 6 |
| 146 | 6 |
| 147 | 6 |
| 148 | 6 |
| 149 | 6 |
| 201 | 1 |
| 205 | 1 |
+-----+-----+
19 rows in set (0.00 sec)
```

2)

```
D4_99207_OM> SELECT country_ID , COUNT(city)
-> FROM locations
-> GROUP BY country_ID;
```

country_ID	COUNT(city)
AU	1
BR	1
CA	2
CH	2
CN	1
DE	1
GB	3
IN	1
IT	2
JP	2
MX	1
NL	1
SG	1
US	4

14 rows in set (0.00 sec)

3)

```
D4_99207_OM> SELECT department_id , AVG(SALARY)
-> FROM employees
-> WHERE commission_pct IS NOT NULL
-> GROUP BY department_id;
```

department_id	AVG(SALARY)
NULL	7000.000000
80	8955.882353

2 rows in set (0.03 sec)

4)

```
107 rows in set (0.00 sec)
-> SELECT JOB_ID,COUNT(employee_ID),SUM(SALARY),MAX(SALARY)-MIN(SALARY)
-> FROM EMPLOYEES
-> GROUP BY JOB_ID;
```

JOB_ID	COUNT(employee_ID)	SUM(SALARY)	MAX(SALARY)-MIN(SALARY)
AC_ACCOUNT	1	8300.00	0.00
AC_MGR	1	12008.00	0.00
AD_ASST	1	4400.00	0.00
AD PRES	1	24000.00	0.00
AD_VP	2	34000.00	0.00
FI_ACCOUNT	5	39600.00	2100.00
FI_MGR	1	12008.00	0.00
HR_REP	1	6500.00	0.00
IT_PROG	5	28800.00	4800.00
MK_MAN	1	13000.00	0.00
MK_REP	1	6000.00	0.00
PR_REP	1	10000.00	0.00
PU_CLERK	5	13900.00	600.00
PU_MAN	1	11000.00	0.00
SA_MAN	5	61000.00	3500.00
SA_REP	30	250500.00	5400.00
SH_CLERK	20	64300.00	1700.00
ST_CLERK	20	55700.00	1500.00
ST_MAN	5	36400.00	2400.00

19 rows in set (0.00 sec)

5)

```
D4_99207_OM> SELECT JOB_ID , AVG(SALARY)
-> FROM employees
-> GROUP BY JOB_ID
-> HAVING AVG(SALARY)>10000;
```

JOB_ID	AVG(SALARY)
AC_MGR	12008.000000
AD PRES	24000.000000
AD_VP	17000.000000
FI_MGR	12008.000000
MK_MAN	13000.000000
PU_MAN	11000.000000
SA_MAN	12200.000000

7 rows in set (0.00 sec)

6)

```
D4_99207_OM>SELECT YEAR(HIRE_DATE) , COUNT(employee_ID) AS employees
-> FROM employees
-> GROUP BY YEAR(HIRE_DATE)
-> HAVING employees > 10;
+-----+-----+
| YEAR(HIRE_DATE) | employees |
+-----+-----+
|          2015 |         29 |
|          2016 |         24 |
|          2017 |         19 |
|          2018 |         11 |
+-----+-----+
4 rows in set (0.00 sec)
```

7)

```
D4_99207_OM> SELECT department_id
-> FROM employees
-> where commission_pct IS NOT NULL
-> GROUP BY department_id
-> HAVING COUNT(employee_ID) > 5;
+-----+
| department_id |
+-----+
|          80 |
+-----+
1 row in set (0.00 sec)
```

8)

```
D4_99207_OM> SELECT employee_ID , COUNT(JOB_ID)
-> FROM job_history
-> GROUP BY employee_ID
-> HAVING COUNT(JOB_ID) > 1;
+-----+-----+
| employee_ID | COUNT(JOB_ID) |
+-----+-----+
|          101 |             2 |
|          176 |             2 |
|          200 |             2 |
+-----+-----+
3 rows in set (0.00 sec)
```

9)

```
D4_99207_OM>SELECT job_id , COUNT(employee_id)
-> FROM job_history
-> WHERE DATEDIFF(end_date, start_date) > 100
-> GROUP BY job_id
-> HAVING COUNT(DISTINCT employee_id) >= 2;
+-----+-----+
| job_id | COUNT(employee_id) |
+-----+-----+
| AC_ACCOUNT |             2 |
| ST_CLERK |             2 |
+-----+-----+
2 rows in set (0.00 sec)
```

10)

```
D4_99207_OM>SELECT department_id, YEAR(hire_date), COUNT(employee_id)
-> FROM employees
-> GROUP BY department_id, YEAR(hire_date)
-> ORDER BY department_id, YEAR(hire_date);
```

department_id	YEAR(hire_date)	COUNT(employee_id)
NULL	2017	1
10	2013	1
20	2014	1
20	2015	1
30	2012	1
30	2013	1
30	2015	2
30	2016	1
30	2017	1
40	2012	1
50	2013	3
50	2014	4
50	2015	12
50	2016	13
50	2017	9
50	2018	4
60	2015	1
60	2016	2
60	2017	2
70	2012	1
80	2014	5
80	2015	10
80	2016	7
80	2017	5
80	2018	7
90	2011	1
90	2013	1
90	2015	1
100	2012	2
100	2015	2
100	2016	1
100	2017	1
110	2012	2

33 rows in set (0.00 sec)

11)

```
D4_99207_OM> SELECT Monthname(HIRE_DATE), COUNT(employee_ID)
-> FROM employees
-> WHERE YEAR(HIRE_DATE) = 2017
-> GROUP BY Monthname(HIRE_DATE);
```

Monthname(HIRE_DATE)	COUNT(employee_ID)
May	2
February	3
December	3
August	1
November	2
January	1
April	1
October	1
March	3
June	2

10 rows in set (0.00 sec)

12)

```
D4_99207_0M>SELECT department_id , MAX(SALARY)
-> FROM employees
-> GROUP BY department_id
-> HAVING MAX(SALARY) > 10000;
```

department_id	MAX(SALARY)
20	13000.00
30	11000.00
80	14000.00
90	24000.00
100	12008.00
110	12008.00

6 rows in set (0.02 sec)